

Table 2 · Task grid and the claim each row tests

Task	Input	Target	Estimator	Metric	Claim tested
Reconstruction	FC	SC	PCA→PLS	demeaned_pearson	cross-modal FC→SC
Reconstruction	SC	FC	PCA→PLS	demeaned_pearson	cross-modal SC→FC
Oracle	FC	FC	BayesianRidge	demeaned_pearson	within-modality ceiling
Oracle	SC	SC	BayesianRidge	demeaned_pearson	within-modality ceiling
Baseline	bv+demo	SC/FC	PCA→PLS	demeaned_pearson	subject-info → connectome
Downstream	bv+demo	Cognition	BayesianRidge	pearson / lift	checkpoint baseline
Downstream	obs_FC	Cognition	BayesianRidge	lift_over_bvdemo	does FC beat baseline?
Downstream	obs_SC	Cognition	BayesianRidge	lift_over_bvdemo	does SC beat baseline?
Downstream	pred_SC / pred_FC	Cognition	BayesianRidge	lift_over_bvdemo	does imputation transfer?
Leak check	any	sex / age	BayesianRidge	bal_acc / pearson	diagnostic, not outcome
Family	pred_SC variants	MZ/DZ/sib	—	AUC	identity signal survives?